

Saddle-node bifurcation of first-passage-time densities induced by a directed shortcut

Xiaoxiao Zhouyi* and Luca Giuggioli
*School of Engineering Mathematics and Technology,
 University of Bristol, Bristol BS8 1TW, United Kingdom*
 (Dated: July 2, 2026)

First-passage-time (FPT) densities in shortcut-assisted search can be bimodal: a fast “direct” arrival and a slow “indirect” one. While this bimodality is well documented, the *morphology* of the density—when a finite-time arrival peak is created or destroyed—has lacked an exact threshold theory. Here we show that, in a minimal model, the morphology transition is an observable-level *saddle-node bifurcation* that can be solved exactly. A directed shortcut to an absorbing target is, after elimination of the target, a rank-one killing defect on a lazy lattice—in the survival sector the directedness enters only the absorption channel—and the diffusive limit is Brownian motion on an interval with an interior δ -sink of strength b at position θ , released at ξ . The FPT density is a *signed* spectral mixture $\Phi(\tau) = \sum_j G_j e^{-\mu_j \tau}$, and its extrema merge at the compact fold criterion $S_1 = S_2 = 0$, where $S_n(\tau) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}$ and $\Phi' = -S_1$, $\Phi'' = S_2$ (primes denoting τ -derivatives). For the source-started convention $\xi = \theta$ the density carries an initial shortcut-capture feature and a finite-time interior peak; numerical continuation shows that the latter survives on a single connected window $0 < b < b_c(\theta)$, annihilated at the computable boundary $b_c(\theta)$ —symmetric about $\theta = \frac{1}{2}$, minimal near $\theta \simeq 0.381$, and diverging as $0.789/\min(\theta, 1-\theta)$ at the endpoints. The fold has the generic saddle-node exponents $(1/2, 3/2)$ with prefactors analytic in the fold data, requires at least three spectral modes, and persists when the walker is released off (but near) the gate, where both features become genuine interior peaks. The local fold is structurally stable for reversible (symmetrizable) low-rank killing, and it is *not* a spectral exceptional point. The classification is validated by exact finite-lattice residues, direct Monte Carlo, an exact channel decomposition, off-gate starts, and Brownian-dynamics simulations with no lattice and no spectral input; it extends to a rank-two cusp and to finite two-dimensional lattices. The result recasts a known phenomenology as a predictive, exactly computable bifurcation of a measurable time-domain observable, testable in feedback-controlled colloidal search and engineered transport networks.

I. INTRODUCTION

The first-passage time (FPT) to a target is a central observable of stochastic transport, search and reaction kinetics [1–5]. Its full distribution $f(t)$, rather than the mean first-passage time (MFPT) alone, controls the timing statistics of diffusion-limited reactions, of intracellular delivery, and of search under deadlines [17], and it is increasingly accessible experimentally through single-particle tracking and feedback-controlled colloidal platforms [13, 14, 18].

A recurring and striking feature of $f(t)$ is *multimodality*. When a searcher has access to a “shortcut”—a relocation channel, a long-range rewired edge, or an active-transport route—the FPT density can become bimodal: fast trajectories that use the shortcut form an early peak, while slow trajectories that explore the domain form a later peak, provided the two time scales are well separated [8]. Random walks on small-world and shortcut-rich networks likewise exhibit shortcut-induced temporal structure in their first-passage statistics [10]. Multimodal capture-time densities are further known for interior partially absorbing traps when several traps are strategically arranged [13, 14], and exact spatiotemporal solutions of confined lattice random walks with defects are by now a

mature program [6, 7].

These works analyze direct/indirect first-passage structure and partial-absorption kinetics [8, 14, 15]—the two-channel (fast-capture vs slow-diffusion) decomposition is explicit already in Ref. [15]; what has been missing is an exactly solvable model whose FPT-density *extrema* undergo a genuine saddle-node fold at a computable shortcut-strength threshold. That is the gap we fill. The distinction matters: the appearance and disappearance of a finite-time arrival peak is a qualitative, controllable property of the distribution that is invisible to the MFPT and to the spectral gap, and is, we argue, a genuine catastrophe of the observable.

Here we supply exactly that classification for a minimal directed shortcut, modeled as a rank-one killing defect on an absorbing lattice. In the survival sector the directed shortcut is equivalent to localized killing: the non-reversibility resides entirely in the absorbing channel, not in the transient propagator (Sec. II). Our contributions are:

1. An *exact solution* (Sec. II): the FPT density is a signed spectral mixture whose amplitudes we derive in closed form from the defect resolvent, both on the finite lattice (Montroll determinant, App. A) and in the diffusive limit (App. B).
2. A *saddle-node fold criterion and computable phase boundary* (Sec. III): the FPT density carries a short-time shortcut-capture feature and a finite-

* xiaoxiao.zhouyi@bristol.ac.uk

time interior peak; the birth and death of the latter is a fold governed by the compact local criterion $S_1 = S_2 = 0$ in signed spectral moments ($\Phi' = -S_1$, $\Phi'' = S_2$), with a single connected window $0 < b < b_c(\theta)$ under the source-started convention. The boundary is symmetric about $\theta = \frac{1}{2}$, minimized near $\theta \simeq 0.381$, and obeys the endpoint law $b_c \sim B^*/\min(\theta, 1 - \theta)$ with $B^* = 0.7890262$ fixed by an explicit half-line transform; a *minimal-mode theorem* (at least three spectral modes) and the generic $(1/2, 3/2)$ saddle-node scaling, with prefactors analytic in the fold data, complete the local classification.

3. An *independent validation chain* (Sec. IV; the exact finite- N checks appear in Secs. II C, III B and App. B, E): exact finite- N residues against the continuum master curve, the fold threshold itself converging as $b_{c,N} - b_c \propto N^{-2.08}$, a direct Monte-Carlo, an exact channel decomposition (the early feature is pure shortcut capture; the late peak is fed by both channels), off-gate starts for which both features are genuine interior peaks, and a Brownian-dynamics realization containing no lattice and no spectral input.
4. *Extensions and scope* (Secs. V, VI): structural stability for reversible (symmetrizable) low-rank killing, a rank-two extension with a triple-peak region and a nondegenerate organizing cusp point, finite-2D evidence with explicit limitations, the demonstration that b_c is *not* a spectral exceptional point, and a quantitative measurability estimate for feedback-controlled colloidal search.

We are careful to separate what is new from what is not. Shortcut-induced bimodality and interior-trap multimodality are known [8, 13, 14]; the lattice Green-function/defect method is that of Refs. [6, 7]; and Ref. [9] concerns bimodality of a splitting probability $P(\omega)$, not of $f(t)$. Our contribution is the exactly solvable *bifurcation-theoretic* account: the fold criterion, the predictive threshold $b_c(\theta)$, and the structural stability of the mechanism.

II. MODEL AND EXACT SOLUTION

A. Lazy ring with a directed shortcut

Consider a lazy random walk on a ring of N sites [Fig. 1(a)]: at each step the walker stays with probability $1 - q$ and hops to each neighbor with probability $q/2$. Site $v = 0$ is an absorbing target. A directed shortcut at a source site u diverts probability $\lambda = \beta(1 - q)$ from the u self-loop onto the directed edge $u \rightarrow v$; upon taking it the walker is absorbed. Removing the absorbed state leaves a transient (sub-stochastic) transition matrix on sites $1, \dots, N - 1$: a Dirichlet path with a rank-one

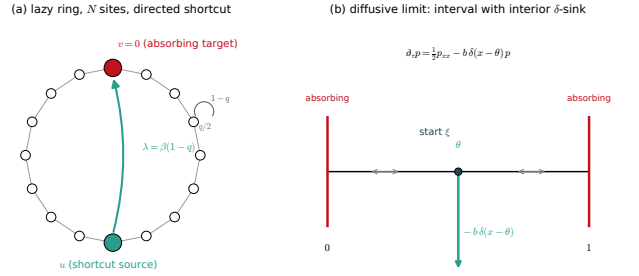


FIG. 1. Model. (a) Lazy random walk on a ring of N sites (stay $1 - q$, hop $q/2$) with an absorbing target $v = 0$ and a directed shortcut $u \rightarrow v$ of rate $\lambda = \beta(1 - q)$; deleting v leaves a rank-one killing defect at u . (b) Diffusive limit: Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink of strength b at θ , started at ξ .

diagonal killing defect of strength $\lambda = \beta(1 - q)$ at u (admissibly $0 \leq \lambda \leq 1 - q$, as it is diverted from the self-loop probability),

$$P_\lambda = P_0 - \lambda |u\rangle\langle u|, \quad (1)$$

so probability leaks irreversibly from u into the target; the rescaled continuum sink strength $b = \lambda N/q$ is introduced below. Although the shortcut is directed, the transient block remains symmetric—the directedness enters only through the (one-way) absorption vector, not the survival dynamics—so its spectrum is real. The eliminated survival operator is therefore self-adjoint: the process is open (sub-Markovian, probability-losing), but the threshold is a time-domain morphology change, not a non-Hermitian spectral (exceptional-point) effect (Sec. VI C).

B. Exact generating function and Montroll determinant

The FPT generating function is rational with an exact Montroll-type determinant. Applying the matrix-determinant lemma to the rank-one defect (1) and evaluating the resulting path-resolvent element in closed form (App. A) gives, with U_m the Chebyshev polynomials of the second kind, $y = 1 + (z^{-1} - 1)/q$ the spectral variable, and $a = q/\lambda$ the dimensionless inverse shortcut strength, the characteristic determinant for a source at u ,

$$D_u(y) = a U_{N-1}(y) + 2 U_{u-1}(y) U_{N-u-1}(y). \quad (2)$$

The FPT probability is then a finite sum of geometric modes,

$$F(t) = \sum_j B_j s_j^{t-1}, \quad s_j = 1 - q + q y_j, \quad (3)$$

where y_j are the roots $D_u(y_j) = 0$ and the residues are $B_j = q \mathcal{N}_{r,u}(y_j)/D'_u(y_j)$ with r the start site and $\mathcal{N}_{r,u}$

the piecewise numerator of App. A. In the antipodal case $u = N/2$ the identity $U_{N-1} = 2U_{N/2-1}T_{N/2}$ factorizes (2) as $D_u = 2U_{N/2-1}(y)[aT_{N/2}(y) + U_{N/2-1}(y)]$ (with T_m the Chebyshev polynomial of the first kind): the bracket carries the sink-coupled spectrum, while the factor $U_{N/2-1}$ carries the antisymmetric (sink-decoupled, node) modes, recovering the reflection-symmetric problem as a special case of the general rank-one theory. Equivalently, in terms of the defected matrix P_λ of (1), the total absorption current has the compact form

$$\tilde{F}_r(z) = z \left[\lambda \langle u | + \frac{q}{2} (\langle 1 | + \langle N-1 |) \right] (I - zP_\lambda)^{-1} |r\rangle, \quad (4)$$

whose two bracketed terms are the shortcut-capture and diffusive-absorption channels resolved in Sec. IV A. In the diffusive scaling the finite mass maps to a continuum density, $F(t) \simeq (q/N^2) \Phi(qt/N^2)$ with $\tau = qt/N^2$.

C. Diffusive limit

Writing $\theta = u/N$ for the fractional shortcut position, $b = \beta(1-q)N/q$ for the rescaled sink strength, $\tau = qt/N^2$ for diffusive time, and writing the transient eigenvalues as $s = 1 - q + q \cos(2w/N)$ —which defines the lattice root variable w , with continuum wavenumber $k = 2w$ and eigenvalue $\mu = k^2/2 = 2w^2$ —the walk converges to Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink [Fig. 1(b)],

$$\partial_\tau p = \frac{1}{2} p_{xx} - b \delta(x - \theta) p, \quad p(0) = p(1) = 0. \quad (5)$$

The finite- b continuum limit is the *weak-per-visit* (partially absorbing) scaling $\lambda = \beta(1-q) = bq/N = O(N^{-1})$: a fixed per-visit shortcut probability would instead approach the absorbing-cut limit $b \rightarrow \infty$. The spectrum of (5) is fixed by the transcendental determinant

$$D_\theta(k; b) = k \sin k + 2b \sin(k\theta) \sin(k(1-\theta)) = 0, \quad (6)$$

with $k = 2w$ and eigenvalues $\mu_j = k_j^2/2 = 2w_j^2$; for $\theta = \frac{1}{2}$ this collapses to $\tan w = -2w/b$. This interval δ -sink (point-interaction) eigenvalue condition is the standard point-sink spectrum, with a long lineage in the localized-sink and imperfect-reactivity literature [11–14, 23]; our exact-solution novelty is not this spectrum but the finite- N directed-shortcut Montroll determinant D_u [Eq. (2)] and the bifurcation classification developed below. The FPT density from a start at $\xi = r/N$ is the signed spectral mixture

$$\Phi_{\xi, \theta}(\tau; b) = \sum_j G_j e^{-\mu_j \tau}, \quad (7)$$

whose amplitudes we derive from the defect resolvent (App. B) as

$$G_{\xi, \theta} = 2w^2 \frac{\phi_{k, \theta}(\xi) I(k, \theta)}{J(k, \theta)}, \quad k = 2w, \quad (8)$$

where

$$\phi_{k, \theta}(x) = \begin{cases} \sin(k(1-\theta)) \sin(kx), & x \leq \theta, \\ \sin(k\theta) \sin(k(1-x)), & x \geq \theta, \end{cases} \quad (9)$$

and $I = \int_0^1 \phi dx$, $J = \int_0^1 \phi^2 dx$ have the closed forms of App. B. Modes with $\sin(n\pi\theta) = 0$ do not couple to the sink and contribute the free Dirichlet term $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish). We have verified (8), the crucial normalization identity $J = \sin(k) D_k/(4b)$, and the full master curve (affected plus node modes) against exact finite- N residues ($N = 1200$) to a relative error $\lesssim 6 \times 10^{-5}$ (per-mode $\sim 2 \times 10^{-4}$) with measured $O(N^{-2})$ convergence (fitted exponent 2.00; App. B, App. E). Table I collects the notation.

III. SADDLE-NODE FOLD CRITERION AND PHASE BOUNDARY

A. Signed spectral moments and the fold criterion

Define the signed spectral moments

$$S_n(\tau; b) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}, \quad (10)$$

so that $\Phi' = -S_1$ and $\Phi'' = S_2$ [differentiate (7)]. We are precise about what folds. Under the source-started convention $\xi = \theta$ the density carries a short-time shortcut-capture feature—a boundary/initial-time contribution whose weight vanishes as $b \rightarrow 0$ —together with *interior* finite-time extrema; an interior extremum is a zero of S_1 . The finite-time interior peak forms a valley–peak pair that is born or annihilated when the two interior extrema merge—a *saddle-node* fold of the density’s critical points,

$$S_1(\tau_c, b_c) = 0, \quad S_2(\tau_c, b_c) = 0, \quad (11)$$

with nondegeneracy $S_3(\tau_c, b_c) \neq 0$ and $\partial_b S_1(\tau_c, b_c) \neq 0$. No positivity is assumed—the G_j are genuinely signed—

TABLE I. Notation.

q	hop probability per step (stay $1 - q$)
β	shortcut fraction of the u self-loop
$\lambda = \beta(1 - q)$	per-step shortcut rate; $a = q/\lambda$
$b = \lambda N/q$	continuum sink strength
$\theta = u/N$, $\xi = r/N$	sink, start; $d = \min(\theta, 1 - \theta)$
$\tau = qt/N^2$	diffusive time
w , $k = 2w$, $\mu = 2w^2$	root, wavenumber, eigenvalue
$B = bd$, $\sigma = \tau/d^2$	boundary layer (App. C)
κ , ℓ , L	trigger rate, gate width, domain size
$L \times L$	2D torus (Sec. V)

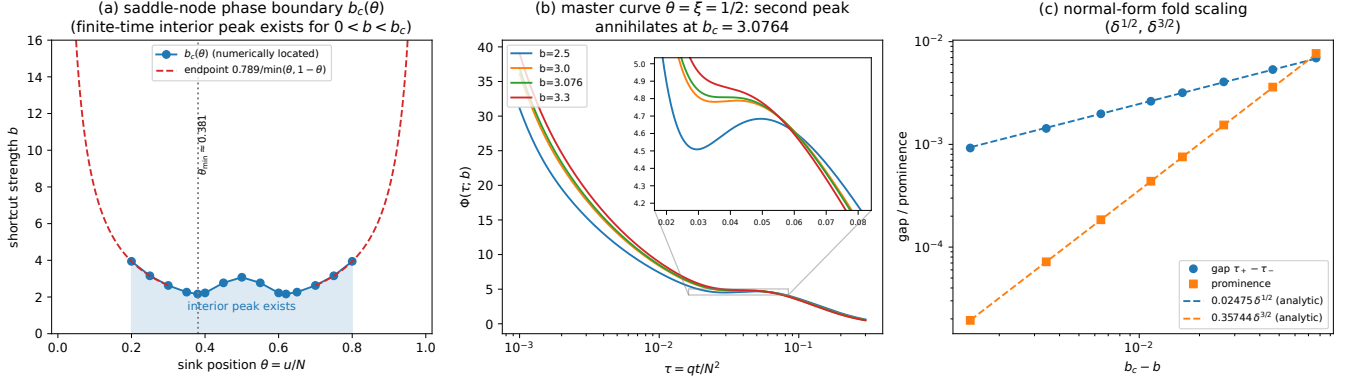


FIG. 2. The exact one-dimensional fold. (a) Phase boundary $b_c(\theta)$ ($\xi = \theta$): the finite-time interior peak exists for $0 < b < b_c(\theta)$; symmetric, minimum at $\theta \simeq 0.381$. Solid points are numerically located over $\theta \in [0.20, 0.50]$ and mirrored by the symmetry $b_c(\theta) = b_c(1 - \theta)$; the dashed curves are the endpoint asymptote $0.789/\min(\theta, 1 - \theta)$ (App. C). (b) Master curve $\Phi(\tau; b)$ at $\theta = \xi = \frac{1}{2}$ for $b = 2.5, 3.0, 3.076, 3.3$, crossing $b_c \simeq 3.076$; the inset magnifies the fold region, where the valley–peak pair merges. (c) Normal-form scaling: gap $\propto (b_c - b)^{1/2}$ and prominence $\propto (b_c - b)^{3/2}$ (fitted exponents 0.498(1), 1.493(2)); dashed lines use the analytic prefactors of Eq. (C2).

so (11) is the correct fold criterion for a signed exponential mixture: the coincidence of a root and a critical point of the generalized (signed) exponential sum S_1 . This places the classification in a classical frame: the FPT density of any finite absorbing chain is a (discrete- or continuous-time) phase-type law [20], whose density is a matrix-exponential—in general signed—mixture, and (11) locates the parameter values at which two extrema of such a mixture merge. The amplitudes G_j , and hence the boundary, depend on the start, so $b_c = b_c(\theta; \xi)$; unless stated otherwise we report $\xi = \theta$. The local fold (11) is exact; the global statement is *numerical*: for $\xi = \theta$, numerical continuation of the interior extrema gives a single connected window $0 < b < b_c(\theta)$ in which the finite-time interior peak exists (the *peak window*), with no re-entrant interval over the scanned range (App. E). The window reaches down to arbitrarily small b : released at the gate, the density diverges as $\Phi \simeq b/\sqrt{2\pi\tau}$ for $\tau \rightarrow 0$ (killing at rate b against the returning heat kernel), while the inherited Dirichlet interior peak survives continuously from $b = 0$, so a valley separates the two features for every sufficiently small $b > 0$. We use theorem language only for the local fold and the exact finite- N algebra.

B. The existence boundary $b_c(\theta)$

Eliminating τ_c between the two equations in (11) yields a scalar boundary $b_c(\theta)$ [Fig. 2(a)]. It is symmetric, $b_c(\theta) = b_c(1 - \theta)$, has a minimum at $\theta \simeq 0.381$ ($b_c \simeq 2.165$), and obeys the endpoint law

$$b_c(\theta) \sim \frac{B^*}{d}, \quad \tau_c \sim c_* d^2, \quad d = \min(\theta, 1 - \theta), \quad (12)$$

where the constant is fixed by the explicit closed-form transform of the half-line boundary-layer prob-

lem [App. C, Eqs. (C3)–(C5)]: $B^* = 0.7890262$ and $c_* = 0.1579221$; interval bisection on the fold independently confirms B^* to six digits ($b_c\theta = 0.789026$ across $\theta \in [0.20, 0.27]$, no reference table) and c_* to three ($\tau_c/d^2 \approx 0.158$); an elementary closed form for B^* remains open. In the antipodal case [Fig. 2(b)]

$$b_c(\tfrac{1}{2}) = 3.07643, \quad \tau_c = 0.038363, \quad (13)$$

independently verified to 10^{-6} , with the numerically fitted near-antipodal expansion $b_c(\frac{1}{2} + \epsilon) \approx 3.0764 - 133\epsilon^2$ (App. C); the finite-lattice threshold converges to this value as $b_{c,N} - b_c \propto N^{-2.08}$ over $N = 100$ –1200 (App. E). The two limits are physically transparent: as $b \rightarrow 0$ the capture feature’s weight vanishes and the density reverts to the single-peaked Dirichlet FPT, while as $b \rightarrow \infty$ the δ -sink becomes an absorbing cut and, for $\xi = \theta$, the density collapses toward $\tau \rightarrow 0$ —so the finite-time interior peak exists only in the finite window $0 < b < b_c$.

C. Normal-form scaling

Near the fold the two merging extrema split with the generic saddle-node exponents

$$\tau_+ - \tau_- \propto (b_c - b)^{1/2}, \quad \Phi(\tau_+) - \Phi(\tau_-) \propto (b_c - b)^{3/2}, \quad (14)$$

which we confirm numerically [Fig. 2(c)], the fitted exponents being 0.498(1) and 1.493(2) (log–log fits over $b_c - b \in [1.4 \times 10^{-3}, 3.6 \times 10^{-2}]$, seven continuation points; App. E). The prefactors are *analytic* in the fold data [App. C, Eq. (C2)]: in the antipodal case $\tau_+ - \tau_- \simeq 0.0247518(b_c - b)^{1/2}$ and prominence $\simeq 0.357444(b_c - b)^{3/2}$, matching directly measured extrema splittings to better than 2×10^{-4} relative. The exponents identify the transition as a fold rather than a smooth

spectral crossover, and—being the generic saddle-node exponents—constitute a system-independent experimental signature (Sec. VI).

D. Minimal mode count

A two-mode fold is impossible, even with signs. Writing $A_j = G_j \mu_j e^{-\mu_j \tau}$, $S_1 = 0$ forces $A_2 = -A_1$, whence $S_2 = (\mu_1 - \mu_2)A_1 = 0$ gives $A_1 = A_2 = 0$, a contradiction. Thus *at least three* modes are required, and the fold ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$ forces *alternating signs* among the three participating contributions. Both statements also follow from Laguerre’s generalization of Descartes’ rule: an n -term real exponential sum has at most as many real zeros, counted with multiplicity, as its coefficient sequence has sign changes [21]—so the double zero of S_1 at a fold requires $n \geq 3$ terms, with alternating signs among the participating triplet. The full fold is realized by a many-mode signed cancellation; at the antipode this coincides with an alternating affected-mode pattern ($G_1 > 0, G_2 < 0, G_3 > 0, \dots$), while away from the antipode the modal signs need not be strictly alternating in eigenvalue order (App. B). Retaining K affected modes in the antipodal case gives $b_c^{(2)}$ (none), $b_c^{(3)} = 3.046$, $b_c^{(4)}$ (none), $b_c^{(5)} = 3.0764$ —the even- K gaps reflecting the odd-mode parity of the alternating structure—converging to (13).

IV. INDEPENDENT VALIDATION

This section assembles the evidence that the fold of Sec. III is a property of the physical process rather than of the spectral representation. The exact finite- N checks—residues against the continuum master curve (Sec. II C, App. B) and the threshold convergence $b_{c,N} \rightarrow b_c$ (App. E)—are complemented by four independent probes (Sec. IV A): a direct Monte-Carlo, an exact channel resolution, non-antipodal robustness, and start-position dependence (under which both features become genuine interior peaks), followed by a Brownian-dynamics realization with no lattice and no spectral input (Sec. IV B).

A. Direct simulation, channel resolution, and start dependence

Four checks confirm that the fold is a physical mechanism rather than an artifact of the spectral representation. (i) *Direct simulation*. A direct Monte-Carlo of the lazy ring, using no spectral input, reproduces the exact FPT density across four decades in τ [Fig. 3(a)], with the simulated splitting probability $\pi_{sc} = 0.56466 \pm 0.00078$ (4×10^5 walkers) agreeing with the exact 0.56522 [Eq. (A6)]. (ii) *Channel resolution*. The absorption current splits exactly into a shortcut entry (λ at u) and diffusive entries ($q/2$ at the neighbors of v), $\Phi = f_{sc} + f_{diff}$

[Fig. 3(b)]: f_{sc} is monotone and carries the short-time mass, while the late finite-time bump receives comparable contributions from both channels (diffusive return and delayed capture). The fold thus annihilates the second arrival-time *scale*, not a single pure channel. (iii) *Non-antipodal robustness*. The interior fold occurs off the reflection axis [Fig. 3(c)]: at $\theta = 0.38$ ($b_c = 2.165$) the interior peak is present at $b = 1.5$, marginal at $b = 2.16$ just below threshold, and absent at $b = 2.8$. (iv) *Start dependence*. Releasing the walker away from the gate ($\xi \neq \theta$) removes the boundary/initial capture feature: both features become genuine *interior* peaks that annihilate through the fold [Fig. 4(a)], and the threshold $b_c(\theta; \xi)$ varies smoothly and symmetrically about $\xi = \theta$ [Fig. 4(b)], so the saddle-node is not an artifact of releasing exactly at the gate. (Far from the gate the morphology changes and the simple peak window is lost.)

B. A direct stochastic realization

Finally, the continuum fold is realized directly by stochastic simulation. A continuum Brownian-dynamics simulation of the delivery-gate protocol (Sec. VI B)—Brownian motion on $[0, 1]$ with absorbing ends and killing at rate b/ℓ inside a narrow gate of width ℓ at θ , integrated by Euler–Maruyama with *no lattice and no spectral input*—reproduces both the exact δ -sink density and the fold: released off the gate ($\xi = 0.44$), it shows two interior peaks below b_c and one above [Fig. 5], matching the exact continuum $\Phi(\tau)$ to histogram precision. Because the simulation contains none of the exact machinery (a finite gate that only coarse-grains to the δ -sink, no lattice, no spectral decomposition), this confirms that the fold is a property of the physical process rather than of the solution method—in particular, not a lattice artifact. The realization is also converged: halving the gate width and refining the Euler–Maruyama step leaves the fold morphology unchanged and the late-peak location stable within the histogram resolution (App. E).

V. EXTENSIONS, STRUCTURAL STABILITY, AND SCOPE

Having validated the one-dimensional fold, we now delimit how far the mechanism extends—and where its present evidence stops.

A. Model-independence

The fold (11) is not special to the ring. For any absorbing continuous-time generator Q_0 (or, analogously, any sub-stochastic transition matrix), a rank- m killing defect $Q_B = Q_0 - UBU^\top$ gives, by the Woodbury identity,

$$\det(sI - Q_B) = \det(sI - Q_0) \det[I_m + BW(s)], \quad (15)$$

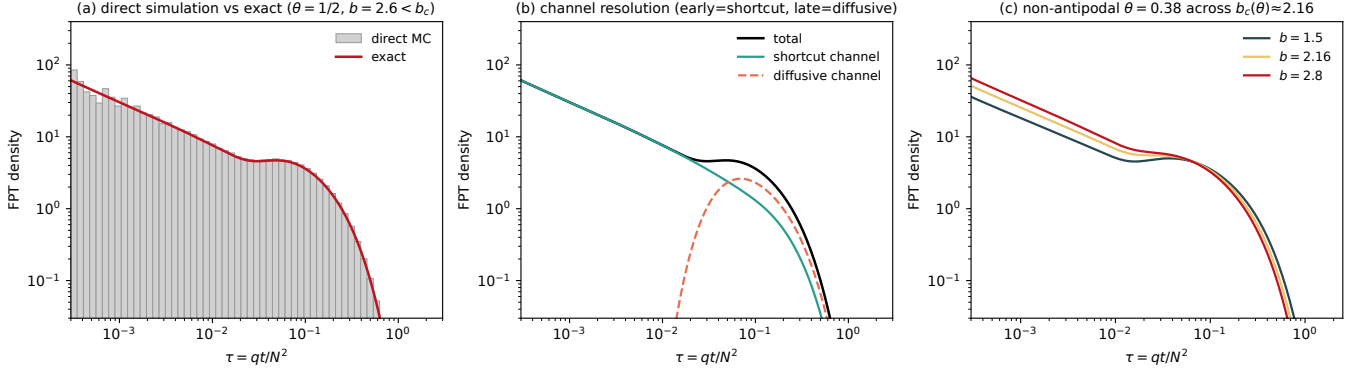


FIG. 3. The fold is a physical mechanism, not an artifact of the exact solution. (a) A direct Monte-Carlo simulation of the lazy ring (grey; 4×10^5 walkers, no spectral input) reproduces the exact FPT density (red) across four decades in τ ; the simulated splitting probability $\pi_{sc} = 0.56466 \pm 0.00078$ agrees with the exact $\pi_{sc} = 0.56522$. (b) Exact channel decomposition $\Phi = f_{sc} + f_{diff}$: the shortcut channel is monotone and carries the short-time mass; the late bump is fed comparably by both channels. (c) The interior fold away from the antipode, $\theta = 0.38$ ($b_c = 2.165$): peak present at $b = 1.5$, marginal at $b = 2.16$, absent at $b = 2.8$. ($N = 100$, $q = 2/3$, $b = 2.6$ in (a,b); gate $u = N/2$ in (a,b), $u = 0.38N$ in (c); start $\xi = \theta$.)

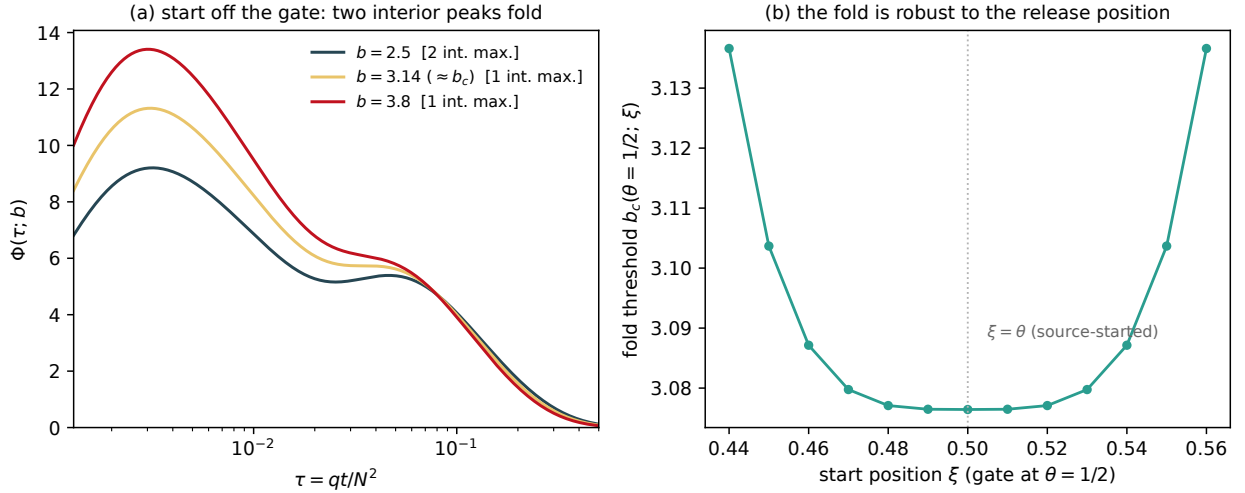


FIG. 4. Start dependence (Sec. IV A). (a) Continuum FPT density $\Phi(\tau; b)$ for a walker released off the gate ($\xi = 0.44$, $\theta = \frac{1}{2}$): both the capture and the late (indirect) features are genuine interior peaks; the late peak and its valley annihilate through the fold as b crosses $b_c \simeq 3.14$ (two interior maxima $\rightarrow$ one). (b) The fold threshold $b_c(\theta = \frac{1}{2}; \xi)$ varies smoothly and symmetrically about the source-started point $\xi = \theta$, so the saddle-node is robust to the release position in a neighborhood of the gate.

with $W_{ab}(s) = \langle u_a | (sI - Q_0)^{-1} | u_b \rangle$. For *reversible* (symmetrizable) transient dynamics with diagonal low-rank killing—as for the symmetric ring block here— Q_B has a real spectrum and the FPT density is again a real signed spectral mixture, whose second-peak birth/death is the fold $S_1 = S_2 = 0$. (For non-reversible directed transient dynamics the determinant identity still holds, but the real modal expansion, and hence the fold classification, requires additional hypotheses.) The Woodbury identity supplies only the finite-rank secular determinant; the universality is the *structural stability* of a generic extrema-pair fold (a codimension-one catastrophe of the scalar density [24]), not a consequence of the identity alone. Accordingly the double-peak region is an *open set* (struc-

turally stable by the implicit-function theorem), and the honest statement is not “every shortcut creates a peak”—false in general—but “whenever a shortcut-induced peak is born or annihilated generically, it does so through a fold of the signed spectral mixture, independent of the underlying geometry.”

B. Multiple shortcuts and the cusp

For two shortcuts at $\theta_1 = \alpha < \theta_2 = \eta$ (positions; β is reserved for the shortcut rate) with strengths b_1, b_2 , the

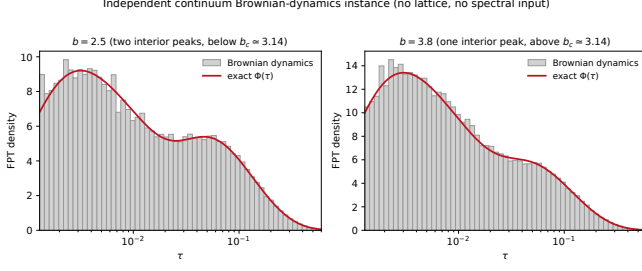


FIG. 5. Direct stochastic realization. A continuum Brownian-dynamics simulation of the delivery-gate protocol (6×10^4 walkers, Euler-Maruyama, gate width $\ell = 0.02$, $\theta = \frac{1}{2}$, start $\xi = 0.44$; no lattice, no spectral input) reproduces the exact δ -sink density $\Phi(\tau)$ (red) and the fold: two interior peaks below $b_c \simeq 3.14$ (left) and one above (right).

secular determinant is (App. D)

$$\begin{aligned} D_{12}(k) = & k \sin k + 2b_1 \sin(k\alpha) \sin(k(1-\alpha)) \\ & + 2b_2 \sin(k\eta) \sin(k(1-\eta)) \\ & + \frac{4b_1 b_2}{k} \sin(k\alpha) \sin(k(\eta-\alpha)) \sin(k(1-\eta)), \end{aligned} \quad (16)$$

the last term being a genuine two-defect interaction that vanishes as the sources coalesce (giving the additive rule $b = b_1 + b_2$). Two shortcuts can produce a *third* FPT peak: at $(\alpha, \eta, b_1, b_2) = (0.38, 0.48, 1.35, 0.14)$, $\xi = 0.463$, the exact ring ($N = 1500$) shows three peaks at $\tau \simeq 3.0 \times 10^{-4}, 5.3 \times 10^{-3}, 6.1 \times 10^{-2}$ [Fig. 6(a)], corresponding to three well-separated arrival-time scales (nominally near-shortcut capture, far-shortcut capture, and diffusive arrival; as for the single-shortcut late peak of Sec. IV A, the later scales are fed by more than one channel). A codimension-two *cusp* point $S_1 = S_2 = S_3 = 0$ —the generic organizing center for a triple-peak region [24]—exists in the same family: solving the three conditions for the three unknowns (η, b_2, τ_c) at fixed $(\alpha, b_1) = (0.38, 1.35)$ locates it at

$$(\alpha, \eta, b_1, b_2, \tau_c) = (0.38, 0.489580512568, 1.35, 1.08389, 0.0012366), \quad (17)$$

with $S_4 \neq 0$ and unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2) = -6.9 \times 10^7 \neq 0$, a genuine codimension-two cusp (App. D). The triple-peak example and the cusp point are *distinct control values in the same rank-two family*—the example demonstrates a three-peak region, while the cusp calculation locates a nearby codimension-two organizing point in the (b_1, b_2) unfolding at fixed geometry (they differ in b_2 , 0.14 vs 1.08, not merely in η). Thus m shortcuts turn the single fold into fold hypersurfaces in b -space, with open regions of two and three peaks and a nondegenerate cusp point as the organizing center [24]; the secular determinants and residues remain finite-rank and explicitly computable, though the catastrophe locations are found numerically.

C. Two dimensions

On an $L \times L$ torus with an absorbing target and a directed shortcut, the same algebra holds [Eq. (15) with the 2D lattice resolvent]. Specifically, we take the lazy walk (stay $1-q$, hop $q/4$ to each of the four neighbors, $q = 2/3$) on the 31×31 torus with absorbing target $v = (0, 0)$, shortcut source $u = (15, 15)$ (antipodal up to lattice rounding, same rate convention $\lambda = \beta(1-q)$), and start $r = (15, 13)$, two sites off the gate—the 2D test therefore realizes the off-gate convention in which both features are genuine interior peaks (cf. Sec. IV A). Numerically, a single diffusive peak at $\beta = 0$ acquires a shortcut-induced capture peak; the two coexist over a window, and the second (late) interior maximum then annihilates by extremum merging near $\beta_c^{2D} \approx 0.68$, its prominence falling to zero between $\beta = 0.68$ and 0.70 [Fig. 6(b,c)]—the same fold morphology as in one dimension. A size sweep locates the threshold at $\beta_c^{2D} = 0.669, 0.684, 0.686, 0.682$ for $L = 21, 31, 41, 51$ (App. E)—nearly size-independent over accessible sizes. One caveat is essential: a single-site sink is marginal in 2D (its self-Green function is logarithmically divergent), so the *bare* threshold is expected to drift as $\beta_c^{2D}(L) \sim 1/\log L$ in the large- L limit; no such drift is yet visible at these sizes, the continuum point-sink limit therefore requires logarithmic renormalization, and we leave a size-independent 2D scaling test (the analogue of (14)) to future work. We thus present the 2D result as finite-lattice evidence that the mechanism is not one-dimensional, not as a proof of 2D universality.

D. Robustness

The fold persists under any perturbation that preserves the nondegeneracy $S_3 \neq 0$, $\partial_b S_1 \neq 0$: to first order $\delta b_c = -(\partial_\epsilon S_1 / \partial_b S_1) \delta \epsilon$ for a perturbation ϵ . Concretely it survives weak hop-rate disorder (which can even help, by lifting accidental dark modes), small shifts of the source or start (Sec. IV A), and the hop probability q , which is already scaled out of the reduced threshold and reduced fold time (the raw step time obeys $t_c \sim N^2 \tau_c / q$). It is destroyed by source coalescence in the rank-two family (rank drop) or by moving the source adjacent to the absorbing boundary. Recall that at $\xi = \theta$ the shortcut-capture feature is a boundary/initial-time contribution, so the interior fold concerns the finite-time interior peak; moving the start off the gate turns the capture feature into an interior peak in its own right (Sec. IV A).

VI. PHYSICAL REALIZATION AND NON-HERMITIAN CHARACTER

A. Embedding

Physically the shortcut is a localized target-delivery channel. From $p'(t) = Q_0 p - b|u\rangle\langle u|p$ the absorption

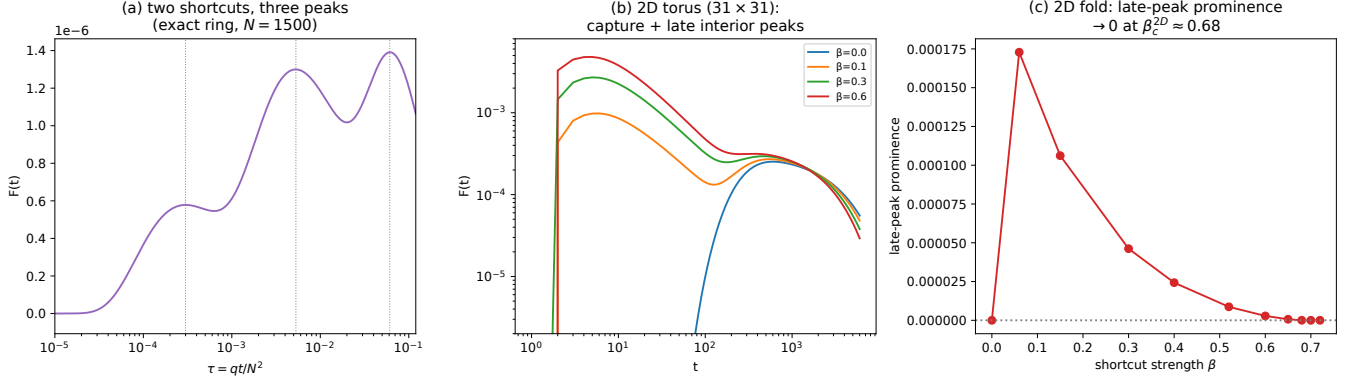


FIG. 6. Extensions. (a) Two shortcuts produce a third peak [exact ring, $N = 1500$; $(\alpha, \eta, b_1, b_2) = (0.38, 0.48, 1.35, 0.14)$, $\xi = 0.463$]. (b) 2D torus $[31 \times 31, q = 2/3$; target $(0, 0)$, source $(15, 15)$, start $(15, 13)$; $\beta = 0, 0.10, 0.30, 0.60$]: capture and late interior peaks. (c) 2D fold: the late-peak prominence vanishes near $\beta_c^{2D} \approx 0.68$ ($L = 31$).

current splits as $f(t) = f_\partial(t) + b p_u(t)$, where f_∂ is the boundary-absorption flux of the *defected* density p (not the $b = 0$ current); while the searcher is at u it is irreversibly delivered to the target at rate b . In the survival sector this is a local killing term $-b|u\rangle\langle u|$, not a transport edge, and a finite gate of physical width ℓ with local trigger rate κ coarse-grains to a δ -sink. Restoring dimensions with domain length L and diffusion constant D in the standard convention $\langle X^2 \rangle = 2Dt$ [the reduced equation (5) has unit variance rate, so $\tau = 2Dt/L^2$, i.e. $t = L^2\tau/(2D)$], the dimensionless sink strength is

$$b = \frac{\kappa \ell L}{2D} \quad (\ell \ll L), \quad (18)$$

so a laboratory quadruple (κ, ℓ, L, D) maps to a definite b ; the smallest threshold $b_c(0.38) \simeq 2.165$ corresponds to a trigger rate $\kappa \simeq 4.3 D/(\ell L)$ at a gate placed at $x \simeq 0.38 L$. The sharp physical statement is that a localized delivery channel supports an *additional arrival-time scale*—a finite-time interior peak in the full FPT density—over the predictable finite window $0 < b < b_c$ of gate strengths, and that this scale is annihilated at b_c through a measurable saddle-node fold: scanning the delivery rate should reveal two arrival-time scales for $b < b_c$, with the late scale merging with its valley and disappearing at b_c .

B. Realizations

We rank candidate realizations by cleanliness. (i) *Feedback-controlled colloidal or active-particle search* is the strongest: a narrow gate at $x = \theta L$ (optical tweezers, a microfluidic pulse, or active-particle feedback) that, upon entry, occasionally delivers the particle to the target and ends the trial; the controlled parameter is the local trigger rate κ —equivalently the dimensionless gate strength $b = \kappa \ell L/(2D)$ (in a frame-based implementation, κ is the per-frame trigger probability divided by the

frame time). Releasing the particle at the gate ($\xi = \theta$, the convention under which $b_c(\theta)$ is quoted), recording first-arrival times over many trials, and scanning b at fixed θ —best near $\theta \simeq 0.38$, where b_c is smallest and the effect largest—should reveal the second peak for $b < b_c(\theta)$ and its annihilation at b_c , with the fold collapse (14) as the sharp signature. A simulated experiment (Brownian dynamics with counting noise) confirms measurability: the robust second-peak observable $R = \Phi(\tau_p)/\Phi(\tau_v)$ (second-peak over valley density) decreases through unity across the fold; $\sim 4 \times 10^4$ delivery-and-terminate trials resolve b_c to roughly ± 0.25 , and $\sim 2 \times 10^5$ to ± 0.1 (App. E); the same holographic-tweezer/feedback stack that realizes controlled first-passage and resetting experiments [18] can implement the delivery gate. We caution that standard resetting-to-origin [18, 19] is *not* this problem: it reinjects probability into the survival domain, whereas the correct protocol is delivery-and-terminate—a gated detection that separates delivery times from bare first-passage times [16]. (ii) *Engineered or small-world transport networks* with a one-way express edge to an absorbing node realize $-b|u\rangle\langle u|$ directly (microfluidic maze with a one-way valve; programmable packet-routing graph; electrical random-walk analog with a tunable shunt to ground; lossy photonic link); two such edges provide the cleanest test of the rank-two triple peak and cusp. (iii) *Intracellular motor capture* to a reaction site realizes it in a coarse-grained, ideally reconstituted, in-vitro setting. In each case the caveat is that only a genuinely one-way, terminating channel maps to the rank-one killing defect.

C. Non-Hermitian character

We stress what the non-Hermitian character is and is not. P_λ is a passive open-Markov loss defect, analogous to an effective non-Hermitian Hamiltonian with a localized absorber (imaginary potential) [25]; it is *not* PT-symmetric, since there is no balanced gain and loss. The

survival-sector operator is itself self-adjoint in the one-defect model (the directedness enters only the absorption vector), so “non-Hermitian” here denotes open-system probability loss, not a spectral mechanism. Crucially, b_c is *not* an exceptional point: an EP requires eigenvalue coalescence with a defective (Jordan) block, whereas we find numerically ($N = 200$) that the eigenvalues remain real, simple, and smooth through b_c , with consecutive gaps bounded away from zero, and the fold arises from cancellation among at least three signed residues, not from spectral degeneracy. The $(b_c - b)^{1/2}$ scaling in (14) is a saddle-node of the scalar $\Phi(\tau; b)$, not EP eigenvalue splitting—the same exponent for a different object. In the reversible diffusive limit the δ -sink operator is self-adjoint (Sturm–Liouville), so no spectral EP exists at all.

VII. DISCUSSION

We do not claim the first observation of shortcut-induced bimodality; the robust result of this paper is the observable-level fold classification with its computable threshold. We have shown that shortcut-induced FPT bimodality, long known phenomenologically, is the observable signature of an exactly solvable saddle-node fold: a rank-one directed defect adds a short-time capture channel, while the pre-existing finite-time interior peak survives, for the source-started convention, only on a single numerically continued interval $0 < b < b_c(\theta; \xi)$, annihilated at b_c through a generic $(1/2, 3/2)$ fold, with a minimal count of three modes, a rank-two extension with a cusp-organized triple-peak region, and the same fold morphology on finite two-dimensional lattices. A direct Monte-Carlo, an exact channel decomposition, and the start-position dependence confirm the mechanism physically. The single most defensible novelty is the bifurcation-theoretic classification with a predictive, computable threshold, distinct from the known bimodality and from the underlying defect-resolvent method. The generic fold exponents and the two-shortcut triple peak are directly measurable, on feedback-controlled colloidal search and engineered transport networks.

Beyond the results verified here, the analysis leaves a small set of open questions. A fully rigorous $N \rightarrow \infty$ statement requires, in addition to the standard convergence of fixed-mode residues (App. B), uniform C^1 convergence of the moment sums $S_{1,2,3}$ near the fold; we state the precise hypotheses and support them by the observed $O(N^{-2})$ convergence, but do not prove uniform C^1 control of the tail. The endpoint constant is fixed by an explicit boundary-layer integral [Eq. (C4)], $B^* = 0.7890262$, and confirmed independently to six digits, though neither it nor the antipodal b_c is known in elementary closed form; and the large- L renormalized scaling of b_c in two dimensions is open. None of these affects the exact finite-system results.

TABLE II. Independent numerical verifications (this work).

Quantity	Agreement
δ -sink jump = D_θ [Eq. (6)]	1.8×10^{-14}
I, J vs. quadrature [Eqs. (B4),(B5)]	$\sim 10^{-9}$
Identity $J = \sin(k)D_k/4b$ [Eq. (B6)]	2.9×10^{-15}
Master curve vs. exact residues ($N = 1200$)	$\lesssim 6 \times 10^{-5}$
$b_{c,N} \rightarrow b_c$, antipodal ($N = 100\text{--}1200$)	$\propto N^{-2.08}$
$b_c(\theta)$ vs. indep. numerics ($\theta \in [0.2, 0.5]$)	$\sim 10^{-6}$
Endpoint $b_c \cdot \theta \rightarrow B^*$ (indep., $\theta \in [0.2, 0.27]$)	0.789026
B^* : half-line (C4) vs. interval bisection	10^{-6}
Fold prefactors: analytic (C2) vs. measured	$\lesssim 2 \times 10^{-4}$
Rank-2 secular $D_{12}(k_j)$ ($N = 800$)	$\lesssim 1.3 \times 10^{-4}$
Cusp residuals $\tau^n S_n / \Phi$ ($n \leq 3$)	$\lesssim 10^{-9}$
MC π_{sc} vs. exact (4×10^5 walkers)	0.6×10^{-3}

ACKNOWLEDGMENTS

We thank colleagues at the Bristol Centre for Complexity Sciences for discussions.

Appendix A: Exact finite- N determinant and residues

We derive Eqs. (2)–(3). Let P_0 be the $(N-1) \times (N-1)$ transient (sub-stochastic) transition matrix of the lazy ring with $v = 0$ deleted; on the interior sites it is a symmetric tridiagonal path with diagonal $1 - q$ and off-diagonal $q/2$. The directed shortcut diverts $\lambda = \beta(1 - q)$ from the self-loop at u into the absorbing target, i.e. it subtracts λ from the (u, u) entry: $P_\lambda = P_0 - \lambda |u\rangle\langle u|$ [Eq. (1)]. The first-passage generating function to v from a start r , $\tilde{F}_r(z) = \sum_{t \geq 1} F(t)z^t$, is a ratio of resolvent elements of P_λ ; by the matrix-determinant lemma its poles are the zeros of

$$\det(I - zP_\lambda) = \det(I - zP_0) [1 + z\lambda G_{uu}^0(z)], \quad (\text{A1})$$

with $G^0(z) = (I - zP_0)^{-1}$. Introducing the spectral variable $y = 1 + (z^{-1} - 1)/q$ maps the lazy path onto the standard second-difference operator, $I - zP_0 = zq(yI - \frac{1}{2}A_{\text{path}})$, with $\det(yI - \frac{1}{2}A_{\text{path}}) = 2^{-(N-1)}U_{N-1}(y)$; by the cofactor structure of the tridiagonal inverse $(yI - \frac{1}{2}A_{\text{path}})^{-1}$,

$$G_{uu}^0(z) = \frac{1}{zq} \frac{2U_{u-1}(y)U_{N-u-1}(y)}{U_{N-1}(y)}. \quad (\text{A2})$$

Substituting (A2) into (A1) and clearing U_{N-1} gives the characteristic determinant $D_u(y) = aU_{N-1}(y) + 2U_{u-1}(y)U_{N-u-1}(y)$ of (2), with $a = q/\lambda$.

The residues follow from the full defect-resolvent column. Writing $A_m := U_{m-1}(y)$ (with the convention $A_0 = U_{-1} = 0$, so $A_N = U_{N-1}$), the Dirichlet-path

Green function has the tridiagonal-inverse entries $\mathcal{G}_{rs} = 2A_{\min(r,s)}A_{N-\max(r,s)}/A_N$, and Sherman–Morrison gives

$$(I - zP_\lambda)^{-1} |r\rangle = \frac{1}{zq} \left[\mathcal{G} |r\rangle - \mathcal{G} |u\rangle \frac{\mathcal{G}_{ur}}{a + \mathcal{G}_{uu}} \right]. \quad (\text{A3})$$

Substituting (A3) into (4) and collecting over the common denominator $a + \mathcal{G}_{uu} = D_u/A_N$ yields, for $r \leq u$,

$$\begin{aligned} \tilde{F}_r(z) = \frac{1}{D_u} & \left[a(A_r + A_{N-r}) + 2A_r A_{N-u} \right. \\ & \left. + \frac{2A_{N-u}}{A_N} (A_{N-r} A_u - A_r A_{N-u}) \right], \quad (\text{A4}) \end{aligned}$$

and the Chebyshev product identity $A_{N-r} A_u - A_{N-u} A_r = A_N A_{u-r}$ [set $y = \cos \varphi$, so $A_m = \sin(m\varphi)/\sin \varphi$, and reduce by product-to-sum] collapses the last term of (A4) to $2A_{N-u} A_{u-r}$: $\tilde{F}_r(z) = \mathcal{N}_{r,u}(y)/D_u(y)$ with the piecewise numerator

$$\mathcal{N}_{r,u} = a[A_r + A_{N-r}] + 2A_{N-u}[A_r + A_{u-r}], \quad r \leq u, \quad (\text{A5})$$

and its mirror ($r \leftrightarrow N - r$, $u \leftrightarrow N - u$) for $r \geq u$. Expanding $\tilde{F}_r = \sum_j B_j z/(1 - s_j z)$ about the pole $z_j = 1/s_j$ and using $dy/dz = -1/(qz^2)$ with $z_j s_j = 1$ converts the residue of $\mathcal{N}_{r,u}/D_u$ into $B_j = q \mathcal{N}_{r,u}(y_j)/D'_u(y_j)$, i.e. the geometric sum (3) with $s_j = 1 - q + q y_j$. Setting $z = 1$ (i.e. $y = 1$, where $A_m(1) = m$) in the shortcut component of (4)—which (A3) reduces to $\mathcal{G}_{ur}/(a + \mathcal{G}_{uu})$ —gives the eventual channel (splitting) probability, the fraction of trajectories absorbed through the shortcut,

$$\pi_{\text{sc}}(r) = \frac{2 \min(r, u) [N - \max(r, u)]}{aN + 2u(N - u)}, \quad (\text{A6})$$

which for the antipodal case $u = N/2$ reduces to $\pi_{\text{sc}} = \rho/(a + N/2)$ with $\rho = \min(r, N - r)$ the start's short-arc distance to the absorbing target. We emphasize that π_{sc} is the eventual splitting probability, *not* the mass under the first peak; conflating the two is a common error we deliberately avoid. The two absorption channels of Sec. IV A have the explicit generating functions

$$\begin{aligned} \tilde{F}_r^{\text{sc}}(z) &= z\lambda \langle u | (I - zP_\lambda)^{-1} | r \rangle, \\ \tilde{F}_r^{\text{diff}}(z) &= z\frac{q}{2} (\langle 1 | + \langle N - 1 |) (I - zP_\lambda)^{-1} | r \rangle, \quad (\text{A7}) \end{aligned}$$

with $\tilde{F}_r = \tilde{F}_r^{\text{sc}} + \tilde{F}_r^{\text{diff}}$ [Eq. (4)]. Their t -domain inverses $f_{\text{sc}}(t) = \lambda \langle u | P_\lambda^{t-1} | r \rangle$ and $f_{\text{diff}}(t) = \frac{q}{2} (\langle 1 | + \langle N - 1 |) P_\lambda^{t-1} | r \rangle$ are the channel densities of Fig. 3(b).

Appendix B: Continuum amplitude from the defect resolvent

We derive Eqs. (8)–(9) from the resolvent of $H_b = -\frac{1}{2}\partial_x^2 + b\delta_\theta$ on $(0, 1)$ with Dirichlet ends, continued to $z = -k^2/2$ (so $\mu = k^2/2 = 2w^2$).

1. *Free resolvent.* With $\varkappa = \sqrt{2z}$ (a spectral variable; distinct from the gate trigger rate κ of Sec. VI A),

$R_0(x, y; z) = 2 \sinh(\varkappa x_{<}) \sinh(\varkappa(1 - x_{>}))/(\varkappa \sinh \varkappa)$; continued to $\varkappa = ik$,

$$r_0(x, y; k) = \frac{2 \sin(kx_{<}) \sin(k(1 - x_{>}))}{k \sin k}. \quad (\text{B1})$$

2. *Defect resolvent (Sherman–Morrison, rank-one δ -sink).*

$$r_b = r_0 - \frac{b r_0(x, \theta) r_0(\theta, y)}{1 + b r_0(\theta, \theta)} = r_0 - \frac{4b \phi(x) \phi(y)}{k \sin k D_\theta}, \quad (\text{B2})$$

with $\phi = \phi_{k,\theta}$ of (9); the pole condition $1 + b r_0(\theta, \theta) = 0$ is exactly $D_\theta(k; b) = 0$ of (6). 3. *Eigenfunction.* The residue eigenfunction is $\psi_j = \phi_{k_j, \theta}$: it is continuous at θ , satisfies the Dirichlet conditions, and obeys the δ -sink jump $\psi'(\theta^+) - \psi'(\theta^-) = 2b\psi(\theta)$. Indeed $\phi'(\theta^+) - \phi'(\theta^-) = -k[\sin(k\theta) \cos(k(1 - \theta)) + \cos(k\theta) \sin(k(1 - \theta))] = -k \sin k$, so the jump condition reads $-k \sin k = 2b \sin(k\theta) \sin(k(1 - \theta))$, which is precisely $D_\theta = 0$ (numerically 1.8×10^{-14}). 4. *Residue = FPT amplitude.* The survival transform is $\bar{Q}_\xi(z) = \int_0^1 r_b(y, \xi; z) dy$ and $\bar{f} = 1 - z\bar{Q}$, so $G_j = \text{Res}_{z=-\mu_j} \bar{f} = \mu_j \text{Res}_{z=-\mu_j} \bar{Q}$. Near a root, $D_\theta(k) \simeq D_k(k_j)(k - k_j)$ with $k - k_j = -(z + \mu_j)/k_j$ (from $z = -k^2/2$), so the pole part of (B2) is $4b \phi(x) \phi(y)/[\sin(k_j) D_k(k_j)(z + \mu_j)]$, and integrating over the first argument gives the normalization-independent amplitude

$$G_j = \frac{4b \mu_j \phi_{k_j, \theta}(\xi) I}{\sin(k_j) D_k(k_j)} = \mu_j \frac{\phi_{k_j, \theta}(\xi) I}{J} = \mu_j \psi_j(\xi) \frac{\int_0^1 \psi_j}{\int_0^1 \psi_j^2}, \quad (\text{B3})$$

i.e. Eq. (8) with $\mu_j = 2w_j^2$; the second equality is the normalization identity (B6) below, which is thus not a mere consistency check but the bridge between the resolvent residue and the eigenfunction-expansion form. 5. *Closed forms.* With $A = \sin k\theta$, $B = \sin k(1 - \theta)$,

$$I(k, \theta) = \frac{1}{k} [B(1 - \cos k\theta) + A(1 - \cos k(1 - \theta))], \quad (\text{B4})$$

$$J(k, \theta) = \frac{1}{2} [\theta B^2 + (1 - \theta) A^2] - \frac{AB \sin k}{2k}, \quad (\text{B5})$$

matching direct quadrature to $\sim 10^{-9}$ (here and below, A, B, C, E are local abbreviations of this appendix, distinct from the residues B_j , the Woodbury matrix B , and the boundary-layer strength $B = bd$). 6. *Normalization identity.* At a root of (6),

$$J(k_j, \theta) = \frac{\sin(k_j) D_k(k_j; b)}{4b}, \quad (\text{B6})$$

with $D_k = \sin k + k \cos k + 2b[\theta \cos(k\theta) \sin(k(1 - \theta)) + (1 - \theta) \sin(k\theta) \cos(k(1 - \theta))]$; this exact algebraic identity (proven below; verified to 2.9×10^{-15}) ties the residue norm to the determinant derivative, and $D_k(k_j) \neq 0$ confirms each affected mode as nondegenerate. The proof is elementary: abbreviate $A = \sin k\theta$, $B = \sin k(1 - \theta)$,

$C = \cos k\theta$, $E = \cos k(1 - \theta)$, so that $\sin k = AE + CB$ and $\cos k = CE - AB$, and use the root condition $k \sin k = -2bAB$ in the form $1/(4b) = -AB/(2k \sin k)$. Then

$$\frac{\sin k D_k}{4b} = -\frac{AB}{2k} (\sin k + k \cos k) + \frac{\sin k}{2} [\theta CB + (1 - \theta)AE],$$

and expanding with $\sin k = AE + CB$, $\cos k = CE - AB$ cancels the mixed $ACBE$ terms, leaving $\frac{1}{2}[\theta B^2(A^2 + C^2) + (1 - \theta)A^2(B^2 + E^2)] = \frac{1}{2}[\theta B^2 + (1 - \theta)A^2]$; hence $\sin k D_k/(4b) = -AB \sin k/(2k) + \frac{1}{2}[\theta B^2 + (1 - \theta)A^2] = J$, Eq. (B5). 7. *Antipodal reduction*. For $\theta = \frac{1}{2}$ ($A = B = \sin w$, $k = 2w$) the root condition reads $\tan w = -2w/b$, whence $\cos w = -b \sin w/(2w)$ and $\sin^2 w = 4w^2/(4w^2 + b^2)$; then $I = \sin w(1 - \cos w)/w$ and $J = \frac{1}{2} \sin^2 w [1 - \sin(2w)/(2w)] = \frac{1}{2} \sin^4 w [1 + b(b + 2)/(4w^2)]$, and (B3) collapses to $G_{\xi, 1/2} = 4w(1 - \cos w) \sin(2w\xi^*)/[\sin^2 w (1 + b(b + 2)/4w^2)]$ with $\xi^* = \min(\xi, 1 - \xi)$. 8. *Node modes*. When $\sin(n\pi\theta) = 0$ the mode decouples from the sink and is the free Dirichlet mode $\sin(n\pi x)$, with amplitude $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish, and all node modes vanish at $\xi = \theta$). The reconstructed master curve (affected plus node) matches exact finite- N residues ($N = 1200$) to $\lesssim 6 \times 10^{-5}$ (worst case in the early- τ tail; per-mode $\sim 2 \times 10^{-4}$) with measured $O(N^{-2})$ convergence (exponent 2.00, median over cases, τ , and N -pairs).

The $N \rightarrow \infty$ limit. Convergence of fixed-mode residues $G_{N,j} \rightarrow G_j$ follows from norm-resolvent convergence of the killed evolution operator under standard hypotheses [22, 23]: (A) $\xi_N \rightarrow \xi$, $\theta_N \rightarrow \theta$; (B) bulk convergence $H_N \rightarrow -\frac{1}{2}\partial_x^2$; (C) δ -sink form scaling $V_{N,\theta} \sim bN$; (D) a symmetrizable transient matrix; (E) simple isolated poles. Passing the fold (11) to the limit additionally requires (F) uniform C^1 convergence of $S_{1,2,3}$ (including ∂_b) on a neighborhood $\tau \in [\tau_c/2, 3\tau_c/2]$ of the fold—not down to $\tau = 0$, where the heat-kernel short-time singularity lives—for which a sufficient practical bound is $\mu_{N,j} \geq cj^2$, $|G_{N,j}| \leq Cj^p$ uniformly. Under (F), fold persistence is a consequence of the implicit-function theorem: the Jacobian $\det DF_c = +S_{1,b}S_3$ is nonzero by nondegeneracy, giving a unique nearby finite- N fold with $b_{c,N} - b_c = O(\varepsilon_N)$, consistent with the directly measured threshold convergence $b_{c,N} - b_c \propto N^{-2.08}$ (App. E). We state (F) as a hypothesis, supported by the numerics, rather than a theorem.

Appendix C: Fold criterion: proofs and asymptotics

Fold criterion. From $\Phi = \sum_j G_j e^{-\mu_j \tau}$, $\Phi' = -\sum_j G_j \mu_j e^{-\mu_j \tau} = -S_1$ and $\Phi'' = \sum_j G_j \mu_j^2 e^{-\mu_j \tau} = S_2$, so an interior extremum is $S_1 = 0$ and two extrema merge at $S_1 = S_2 = 0$, Eq. (11). The parameter derivative of a root of (6) is $\partial_b k_j = -D_b/D_k$ with

$D_b = 2 \sin(k\theta) \sin(k(1 - \theta))$ and D_k as in (B6), which propagates into $\partial_b S_1$ and gives the nondegeneracy condition needed for a structurally stable fold. In practice $\partial_b S_1$ is evaluated by high-precision central differences (mpmath, 40 digits), with $\partial_b k_j = -D_b/D_k$ serving as an independent consistency propagation.

Normal-form prefactors. Expanding S_1 about the fold and using $S_2(\tau_c, b_c) = 0$,

$$S_1(\tau, b) \simeq \frac{1}{2} S_3 (\tau - \tau_c)^2 - S_{1,b} (b_c - b), \quad (C1)$$

with $S_{1,b} = \partial_b S_1 > 0$ and $S_3 > 0$ here, so the two extrema sit at $\tau_{\pm} - \tau_c = \pm \delta$ with $\delta = [2S_{1,b}(b_c - b)/S_3]^{1/2}$, and the prominence is $\Delta\Phi = \Phi(\tau_+) - \Phi(\tau_-) = -\int_{\tau_-}^{\tau_+} S_1 d\tau = \frac{2}{3} S_3 \delta^3$. The scaling amplitudes in (14) are therefore *analytic* in the fold data,

$$\tau_+ - \tau_- = 2 \left[\frac{2S_{1,b}}{S_3} \right]^{1/2} (b_c - b)^{1/2},$$

$$\Delta\Phi = \frac{4\sqrt{2}}{3} \frac{S_{1,b}^{3/2}}{S_3^{1/2}} (b_c - b)^{3/2}, \quad (C2)$$

giving at the antipodal fold $0.0247518(b_c - b)^{1/2}$ and $0.357444(b_c - b)^{3/2}$, which match directly measured extrema splittings to better than 2×10^{-4} relative for $b_c - b \leq 10^{-3}$.

Minimal-mode theorem. As in Sec. IIID, $S_1 = S_2 = 0$ has no two-mode solution with $A_j \neq 0$; the minimal count is three, with the alternating-sign ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$. This retires an earlier (incorrect) two-mode heuristic and shows the fold is intrinsically a many-mode cancellation.

Endpoint law. As $d = \min(\theta, 1 - \theta) \rightarrow 0$ the gate approaches the absorbing boundary. Rescaling $y = x/d$, $\sigma = \tau/d^2$, $B = bd$ (the boundary-layer strength; not the residues B_j of App. A) and letting $d \rightarrow 0$ turns (5) into a half-line boundary-layer problem: diffusion on $(0, \infty)$ with an absorbing end at 0 and a δ -sink of strength B at $y = 1$, started at $y = 1$ (the far boundary recedes as $1/d$). This problem is exactly solvable. With $\varkappa = \sqrt{2s}$ the half-line Dirichlet resolvent is $r_0^{\text{hl}}(y, y'; s) = 2 \sinh(\varkappa y_{<}) e^{-\varkappa y_{>}} / \varkappa$, and Sherman-Morrison [as in (B2)] gives the arrival-time transform (boundary flux plus sink capture) in closed form,

$$\hat{f}(s; B) = \frac{e^{-\varkappa} + (B/\varkappa)(1 - e^{-2\varkappa})}{1 + (B/\varkappa)(1 - e^{-2\varkappa})}, \quad \varkappa = \sqrt{2s}, \quad (C3)$$

which obeys $\hat{f}(0) = 1$ (certain absorption) and reduces at $B = 0$ to the Lévy transform $e^{-\varkappa}$. The half-line operator has purely continuous spectrum, so the discrete criterion $S_1 = S_2 = 0$ is replaced by its density form $\Phi'_{\text{hl}} = \Phi''_{\text{hl}} = 0$: Bromwich inversion along the branch cut $s = -v^2/2$ ($\varkappa = iv$) gives the real integral representation

$$\Phi_{\text{hl}}(\sigma; B) = \frac{1}{\pi} \int_0^\infty \frac{v e^{-\sigma v^2/2} (1 + g) \sin v}{1 + 2g \cos v + g^2} dv, \quad (C4)$$

with $g(v) = 2B \sin(v)/v$ [the denominator equals $(g + \cos v)^2 + \sin^2 v$, strictly positive on the support]. Imposing the fold on (C4) selects

$$B^* = 0.7890262, \quad \sigma^* = c_* = 0.1579221, \quad (\text{C5})$$

whence $b_c(\theta) \sim B^*/d$ and $\tau_c \sim c_* d^2$ [Eq. (12)]: the endpoint constant is fixed by an explicit one-dimensional integral, though an elementary closed form remains open. Independently, resolving $b_c(\theta)$ on the *interval* by bisection on the fold count—with the root search extended to the high- k fold mode ($w \sim 1.78/d$), which a naive locator misses—gives $b_c(\theta)\theta = 0.789026$ stably for $\theta \in [0.20, 0.27]$, agreeing with (C5) to six digits without any reference table (below $\theta \simeq 0.15$ the fold mode is too high- k for clean interval resolution). Near the antipode the boundary is smooth and even in ϵ ; a quadratic fit to the numerically continued threshold over $|\epsilon| \leq 0.04$ gives $b_c(\frac{1}{2} + \epsilon) \approx 3.0764 - 133\epsilon^2$ (fitted coefficient -133.1 , with the quartic term controlled), with the antipodal values (13) from $S_1 = S_2 = 0$ directly ($S_3 = 2.8286 \times 10^5$, $\partial_b S_1 = 21.662$, nondegenerate). The truncated thresholds ($K = 2$ none, $K = 3 \rightarrow 3.046$, $K = 4$ none, $K = 5 \rightarrow 3.076$) confirm rapid, parity-structured convergence.

Appendix D: Two-shortcut determinant and cusp

For two sources the secular condition is $\det[I_2 + BW(k)] = 0$ with $W_{ab} = r_0(k; \theta_a, \theta_b)$ the interval resolvent (B1); expanding the 2×2 determinant gives $D_{12}(k)$ of (16), whose cross term $\propto b_1 b_2$ is the genuine two-defect interaction (it $\rightarrow 0$ as $\alpha \rightarrow \eta$, recovering a single defect of strength $b_1 + b_2$); the exact ring ($N = 800$) eigenvalues satisfy $D_{12}(k_j) = 0$ to $\lesssim 1.3 \times 10^{-4}$ (Table II). The rank-two residue eigenfunction follows from the Woodbury pole structure: at a root of $\det[I_2 + BW]$ the resolvent residue factorizes on the null vector $c^{(j)} \in \ker[I_2 + W(k_j)B]$, giving $\psi_j(x) = -\sum_a b_a c_a^{(j)} r_0(k_j; x, \theta_a)$, and $G_j = \mu_j \psi_j(\xi) \int \psi_j / \int \psi_j^2$ exactly as in the rank-one case, with $\int \psi_j = -\sum_a b_a c_a I_{\theta_a}$ and $\int \psi_j^2 = \sum_{ab} b_a c_a b_b c_b C_{ab}$. The constituents are I_θ [Eq. (B4)] and the overlap $C_{ab} = \int_0^1 r_0(x, \theta_a) r_0(x, \theta_b) dx$, which evaluates in closed form by splitting the integral at $\theta_a \leq \theta_b$ into three segments of sine products (the explicit three-line expression is displayed in the accompanying script `dpma_cusp_verify.py`). The cusp solves the three conditions $S_1 = S_2 = S_3 = 0$ for the three unknowns (η, b_2, τ_c) at fixed $(\alpha, b_1) = (0.38, 1.35)$ —hence the many digits quoted in (17)—with $S_4 \neq 0$ and a nonzero unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2)$. At the located point (17) an independent reimplementations gives normalized residuals $\tau S_1/\Phi \approx 7 \times 10^{-12}$, $\tau^2 S_2/\Phi \approx 5 \times 10^{-11}$, $\tau^3 S_3/\Phi \approx 4 \times 10^{-10}$, $\tau^4 S_4/\Phi \approx -1.67$, and $\det \partial(S_1, S_2)/\partial(b_1, b_2) \approx -6.9 \times 10^7$, confirming a nondegenerate codimension-two cusp.

Appendix E: Numerical methods and reproducibility

Exact finite- N FPT poles are computed from the symmetric transient block via `numpy eigvalsh`, with amplitudes from the closed-form residues; high-precision cross-checks use `mpmath`. Continuum spectra are the roots of (6)/(16), and amplitudes use (8) and App. D. All scripts are deterministic. Table II collects the verification tolerances quoted throughout. The figures are generated by `dpma_prr_figures.py` (fold results and extensions) and `dpma_schematic.py` (model); the existence boundary by `dpma_saddle_node_bc_theta.py`; the amplitude/master curve by `dpma_general_u_master_curve.py`; the fold scaling by `dpma_saddle_node_certification.py`; the cusp by `dpma_cusp_verify.py`; the rank-two determinant and triple peak by `dpma_multishortcut.py`; the 2D test by `dpma_2d_finite_lattice.py`; the direct-simulation, channel-resolution, and non-antipodal panels by `dpma_channel_mc.py`; the start-dependence panels by `dpma_start_dependence.py`; the Brownian-dynamics realization by `dpma_brownian_fold.py`; the endpoint-constant resolution by `dpma_endpoint_low.py`; the half-line transform inversion and fold [Eqs. (C3)–(C5)] by `dpma_halfline_bstar.py`; the normal-form prefactors and the near-antipodal quadratic coefficient by `dpma_normal_form.py`; the window-connectivity scan by `dpma_window_scan.py`; the 2D size sweep by `dpma_2d_lsweep.py`; the finite- N threshold convergence by `dpma_bcN_convergence.py`; the Brownian-dynamics convergence sweeps by `dpma_bd_convergence.py`; and the measurability estimate by `dpma_measurement.py`. All randomized computations use fixed seeds recorded in the scripts. The 1D boundary $b_c(\theta)$ is obtained by bisection on the valley-peak-pair count over $\theta \in [0.2, 0.5]$, $b \in [0.5, 20]$, retaining the affected roots with $w \leq 40$ (i.e. $k \leq 80$). Connectivity of the window is verified separately by a dense- b scan [$\Delta b = 0.02$, seven values $\theta \in \{0.20, 0.25, \dots, 0.50\}$, spectrum deepened to $w \leq 120$ so the mode sum is converged down to the small fold times of off-center θ]: exactly one transition per θ —two interior extrema for all $0 < b < b_c$, none above—with no re-entrance (`dpma_window_scan.py`); the $b \rightarrow 0$ end is closed analytically by the short-time argument of Sec. III A, and the tabulated boundary agrees with an independent reimplementations to $\sim 10^{-6}$ (Table II). The normal-form exponents are log-log fits over $b_c - b \in [1.4 \times 10^{-3}, 3.6 \times 10^{-2}]$ (seven continuation points), giving 0.498(1) and 1.493(2) (OLS standard errors). The 2D fold is computed at $L = 31$ (target (0, 0), source (15, 15), start (15, 13), horizon $T = 6000$ steps) by extremum merging (the second interior maximum's prominence $\rightarrow 0$ near $\beta \approx 0.68$), not an ad-hoc valley heuristic; peak detection separates capture from late arrivals at $t = 120$ and uses a relative height threshold $10^{-3} \max F$, with the fold location insensitive to both over wide ranges. A size sweep with the same criterion gives $\beta_c^{2D} = 0.669, 0.684, 0.686, 0.682$ at $L = 21, 31, 41, 51$

(horizon scaled as L^2 ; `dpma_2d_Lsweep.py`).

Threshold and discretization convergence. Bisection on the exact ring gives the finite-lattice fold threshold converging monotonically from above to the continuum value, $b_{c,N}(\frac{1}{2}) - b_c = 4.7 \times 10^{-3}, \dots, 2.6 \times 10^{-5}$ over $N = 100$ –1200, with fitted $|b_{c,N} - b_c| \propto N^{-2.08}$. The Brownian-dynamics realization is likewise converged: halving the gate width ($\ell = 0.04, 0.02, 0.01$) and halving/doubling the Euler–Maruyama step (3×10^{-5} to 7.5×10^{-6}) leaves the late-peak location within two histogram bins of the exact τ_p in every cell, while the ratio observable agrees with the exact δ -sink value at the 1–3.5% level (within ~ 2.6 standard errors; a small first-order killing bias shrinks with the step), so the fold morphology of Fig. 5 is not a discretization artifact. The Monte-Carlo uses 4×10^5 vectorized walkers with a fixed seed, logarithmic τ -binning, and shortcut-vs-diffusive absorption events recorded per walker; the quoted π_{sc} uncertainty is the binomial standard error, and the exact π_{sc} is evaluated from the closed form (A6). The histogram of Fig. 3(a) uses 59 logarithmic bins with a median occupancy of $\sim 4 \times 10^3$ events; occupancy falls to $O(1)$ only in the extreme late- τ tail, where the visible Poisson scatter remains consistent with the exact curve.

Measurability. The simulated experiment behind Sec. VIB draws $N_t = 4000$ delivery-and-terminate trials per run from the continuum gate protocol of Fig. 5 ($\ell = 0.02$, $\theta = \frac{1}{2}$, $\xi = 0.44$, Euler–Maruyama step 1.5×10^{-5}) and evaluates the ratio observable $R = \hat{\Phi}(\tau_p)/\hat{\Phi}(\tau_v)$ from histogram densities in fixed $\pm 20\%$ windows around the exact below-threshold peak and valley locations ($\tau_p = 0.0462$, $\tau_v = 0.0253$); the exact $R(b)$ decreases monotonically through unity across the fold, and the simulated estimates track it within counting noise. Ten independent runs give $\text{s.d.}(R) \simeq 0.061$ at $N_t = 4000$; the exact $R(b)$ table gives the local sensitivity $|dR/db| \simeq 0.09$ per unit b near the fold, so with $\text{s.d.}(R) \propto N_t^{-1/2}$, $N_t \simeq 4 \times 10^4$ trials resolve b_c to roughly ± 0.25 and $N_t \simeq 2 \times 10^5$ to ± 0.1 (`dpma_measurement.py`).

Code and data availability. All scripts are deterministic; the complete code generating every figure and table, together with the simulation histogram data underlying Figs. 3–5, is available from the authors upon reasonable request and will be deposited in a public repository upon publication.

-
- [1] S. Redner, *A Guide to First-Passage Processes* (Cambridge University Press, Cambridge, 2001).
 - [2] R. Metzler, G. Oshanin, and S. Redner (eds.), *First-Passage Phenomena and Their Applications* (World Scientific, Singapore, 2014).
 - [3] O. Bénichou, C. Loverdo, M. Moreau, and R. Voituriez, *Intermittent search strategies*, Rev. Mod. Phys. **83**, 81 (2011).
 - [4] S. Condamin, O. Bénichou, V. Tejedor, R. Voituriez, and J. Klafter, *First-passage times in complex scale-invariant media*, Nature (London) **450**, 77 (2007).
 - [5] O. Bénichou and R. Voituriez, *From first-passage times of random walks in confinement to geometry-controlled kinetics*, Phys. Rep. **539**, 225 (2014).
 - [6] E. W. Montroll and G. H. Weiss, *Random walks on lattices. II*, J. Math. Phys. **6**, 167 (1965).
 - [7] L. Giuggioli, *Exact spatiotemporal dynamics of confined lattice random walks in arbitrary dimensions: A century after Smoluchowski and Pólya*, Phys. Rev. X **10**, 021045 (2020).
 - [8] A. Godec and R. Metzler, *Universal Proximity Effect in Target Search Kinetics in the Few-Encounter Limit*, Phys. Rev. X **6**, 041037 (2016).
 - [9] T. G. Mattos, C. Mejía-Monasterio, R. Metzler, and G. Oshanin, *First passages in bounded domains: When is the mean first passage time meaningful?*, Phys. Rev. E **86**, 031143 (2012).
 - [10] E. Almaas, R. V. Kulkarni, and D. Stroud, *Scaling properties of random walks on small-world networks*, Phys. Rev. E **68**, 056105 (2003).
 - [11] G. Wilemski and M. Fixman, *General theory of diffusion-controlled reactions*, J. Chem. Phys. **58**, 4009 (1973).
 - [12] A. Szabo, K. Schulten, and Z. Schulten, *First passage time approach to diffusion controlled reactions*, J. Chem. Phys. **72**, 4350 (1980).
 - [13] D. S. Grebenkov, *Paradigm Shift in Diffusion-Mediated Surface Phenomena*, Phys. Rev. Lett. **125**, 078102 (2020).
 - [14] P. C. Bressloff, *Diffusion-mediated absorption by partially-reactive targets: Brownian functionals and generalized propagators*, J. Phys. A **55**, 205001 (2022).
 - [15] A. Godec and R. Metzler, *First passage time statistics for two-channel diffusion*, J. Phys. A: Math. Theor. **50**, 084001 (2017).
 - [16] A. Kumar, Y. Scher, S. Reuveni, and M. S. Santhanam, *Inference from gated first-passage times*, Phys. Rev. Research **5**, L032043 (2023).
 - [17] D. S. Grebenkov, R. Metzler, and G. Oshanin, *Strong defocusing of molecular reaction times results from an interplay of geometry and reaction control*, Commun. Chem. **1**, 96 (2018).
 - [18] O. Tal-Friedman, A. Pal, A. Sekhon, S. Reuveni, and Y. Roichman, *Experimental Realization of Diffusion with Stochastic Resetting*, J. Phys. Chem. Lett. **11**, 7350 (2020).
 - [19] A. Pal and S. Reuveni, *First Passage under Restart*, Phys. Rev. Lett. **118**, 030603 (2017).
 - [20] M. F. Neuts, *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach* (Johns Hopkins University Press, Baltimore, 1981).
 - [21] G. Pólya and G. Szegő, *Problems and Theorems in Analysis II* (Springer, Berlin, 1976), Part V.
 - [22] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed. (Springer, Berlin, 1995).
 - [23] S. Albeverio, F. Gesztesy, R. Høegh-Krohn, and H. Holden, *Solvable Models in Quantum Mechanics*, 2nd ed. (AMS Chelsea, Providence, 2005).
 - [24] V. I. Arnol'd, *Catastrophe Theory*, 3rd ed. (Springer, Berlin, 1992).
 - [25] Y. Ashida, Z. Gong, and M. Ueda, *Non-Hermitian physics*, Adv. Phys. **69**, 249 (2020).